Full volume from shared-radius certificates

Original source and complete geometry

Fixed certified simple-white plat; prescribed knot-producing parity.
Exact shared-face coefficient, marked zeros, both level parities.

$$\lim_{N\to\infty} \frac{\log Z_{D,N}}{N} \quad = \quad \max_{q\in\mathcal{Q}_D} \sum_e h(q_e) \quad = \quad \frac{\mathrm{Vol}(Y_D)}{2\pi}$$

Entire CROSS filling period: proved

Nonzero geometric saddle; all original Fourier charges and
whole-domain endpoint currents controlled.

$$\lim_{N\to\infty} \frac{2\pi}{N} \log |I_{\mathrm{cross},N}| \quad = \quad \mathrm{Vol}(S^3 \setminus K_D(\mathbf{p}))$$

Finite covers of genuine source sectors

Give each branch one common face vector, with protected zeros.
Cover every non-CROSS string and verify a strict bound:

$$G_B(L^B) \quad < \quad \Phi_D \quad \text{for every branch } B.$$

Forty branches pass where a simpler anchored test cannot.
Complete-packing balance and the $2R$ tests remain sufficient.

Full original invariant: proved under the certificate

The LEFT/RIGHT remainder has a strict global exponential gap.

$$\lim_{N\to\infty} \frac{2\pi}{N} \log |K_N(K_D(\mathbf{p}))| \quad = \quad \mathrm{Vol}(S^3 \setminus K_D(\mathbf{p}))$$

Every independently long fixed signed twist vector;
threshold depends on D. Twists are fixed before $N \to \infty$.

Proved family at EVERY fixed bridge width $k \geq 2$

A separate chain at every initial bridge gap of the core.
$3k - 3 + 2\sum_j m_j$ independent regions, for every fixed depth vector.

Exponents divisible by 3 give minimal bridge number exactly k.

Open: extend to general plat diagrams

The finite shared-radius theorem applies wherever its tests hold.
A strict bound for every genuine sector remains to be proved.